# Lakshyam: progress since M1

*A browser life simulation where a Class 11–12 student lives one life from age 11 to 36 and feels what money choices do over time.*

## 1. Current approach and changes since M1

The core approach is unchanged. It is still a seeded, pure-JavaScript engine that reads JSON content and drives React screens, with the same 24-lesson curriculum (L01–L24) across planning and budgeting, saving, investing and staying safe. It also keeps both guardrails: *educate, do not recommend*, and *simulations are assumptions, not promises*. What changed is how the lessons reach the player. In our own playthroughs the M1 card-per-year loop felt like a quiz, so we moved the main path to a **story mode** the player walks through. The original year-by-year card mode is kept and still covers all 24 lessons.

| Change | What it is now | Why |
|---|---|---|
| Story mode (new) | 6 missions (prologue, school, college, first job, family, dream) with 36 scenes, 97 choices and 32 outcomes. Each mission has objectives, a flowchart of where the player's choices led, and an evidence page at the end. | A story with stakes holds attention better than isolated cards. |
| STOP warnings and timers | 16 scenes flash a STOP warning before a risky step (OTP sharing, “guaranteed double”, renting a bank account). One scene runs on a countdown, like a real scam call. | Scams work by rushing people; the game rehearses slowing down. |
| Walkable world | A pixel village with Telugu-style houses, shops, a bank, ambient life and NPCs, plus a separate town map for the first-job chapter. | Familiar places; lessons happen where they would. |
| Mini-games | **Interest Race**: to win the school scholarship the player compares three offers by computing P×R×T/100, 4 of 5 right to pass. **Savings Snowball**, **Jar Dash** and **Scam Smash** are in an arcade. All can also be played from the title screen. | Practise the formula and the reflex, not just read about them. |
| Phone app | In-game bank, savings, a practice stock market with fake prices, and a currency exchange. | Investing taught without recommending any real product. |
| Languages | English, Hindi and Telugu are complete: 969 interface strings in each, plus all story, lesson, glossary and FAQ content. The language can be switched at any time. | The brief asks for Indian-language readiness; we chose to ship it. |
| Learning support | **Gyaan** pages (24 lesson pages and 12 topics), a glossary of 79 terms, a 56-question FAQ, 6 documents to inspect, and “Ask Ammamma”, an offline mentor that answers from the FAQ. The optional AI mentor stays server-side. | Works in a classroom with no internet and no API key. |
| Polish | A storybook intro, character looks that change with life stage, a redesigned budget (jars) screen, a dream picker, title and hero-picker screens, and background music with a mute toggle. | An intuitive interface was an explicit judging criterion. |

The other M1 mechanics (echoes, Ammamma's glasses, documents, Sahukar's chair, Parallel Life replay) are built and tested in year mode; only part of that set is in story mode so far. The engine contract did not change, so content, engine and UI kept working in parallel.

## 2. Experiments run so far and results

### Experiment A: 900 automated story-mode lives

We wrote a headless simulator that plays the real engine with three policies, 300 lives each with different seeds. *Good* always picks the best-quality choice, *bad* always picks the worst, and *random* picks uniformly. It checks that every life finishes, that no state gets stuck, and that money is conserved (every rupee in or out has a source), then records the ending, net worth and lessons.

| Metric (median unless noted) | Good | Random | Bad |
|---|---|---|---|
| Errors / stuck lives / money-conservation failures | 0 / 0 / 0 | 0 / 0 / 0 | 0 / 0 / 0 |
| Mission outcome | best in all 6 | mixed (12 jail, 18 bankrupt, 4 dream) | worst in all 6 |
| Lessons seen (of 24), minimum | 19 | 18 | 21 |
| Lessons applied correctly | 17 | 8 | 0 |
| Final net worth | ₹6,373 | ₹116,496 | ₹88,605 |
| Final debt | ₹0 | n/a | ₹156,570 |
| Money ending (count) | secure, no dream (300) | secure 268, dream on time 22, close 10 | secure, no dream (300) |

*Table 1. Headless runs of the story engine (300 seeded lives per policy, about 22 s in total).*

**What it showed.** The story branching works: choice quality maps cleanly onto mission outcomes (all best vs all worst), and lessons *applied* fall from 17 to 0 as play gets worse, which is exactly the signal a teacher report needs. The engine is robust: 0 crashes, 0 stuck

lives and exact money conservation. It also found **two real problems** we would not have seen by hand:

- **The money ending does not reward good play.** Careful players end with a small net worth (₹6,373), no emergency fund and no dream reached, while reckless players carry more assets alongside large debt. The ending id cannot tell them apart. Our working explanation: story choices pay out mainly in lessons and outcomes, while the money ending still uses year-mode thresholds.
- **Story-mode coverage is 18–21 of 24 lessons, not 24 of 24.** Year mode guarantees full coverage; story mode skips a few lessons on some paths.

### Experiment B: content validation and rule checks

A content validator checks all story, card, lesson, glossary and translation files against the spec: required fields, lesson ids, age windows, links between scenes, and that every string exists in all 3 languages. It now reports **0 errors and 0 warnings** over 105 village cards, 36 scenes, 24 lessons and 79 glossary terms.

### Experiment C: internal playtests

Team playthroughs found that the first Interest Race screen was confusing (we added a worked example before round 1) and that the pixel font made “2” look like “8” (money now uses the body font). These are anecdotal; a pre/post quiz is the planned measure.

## 3. Progress on testing

| Layer | What runs | Status |
|---|---|---|
| Engine unit tests | 80 Node tests: seeded determinism, budget and EMI rules, lessons inside their age windows, glasses, documents, echoes, Sahukar's chair, replay, endings, money conservation, and story-mode lives, timers and flowcharts. | 80 / 80 pass |
| Content validator | Schema, cross-references and 3-language completeness (969 UI strings × 3, all translated). | 0 errors, 0 warnings |
| End-to-end (Playwright) | 8 browser scenarios: prologue to flowchart, STOP warning, walking with arrow keys, language switch, save and continue, offline Ask Ammamma, Interest Race, town jump. Layout checks at 1024×768, 1280×720 and 1366×768 laptop sizes. | 11 / 11 pass |
| Simulation (Exp. A) | 900 lives across 3 policies with invariant checks. | Passes; 2 balance issues logged |
| Build | Production Vite build; the game runs fully offline as a static site. | Passes |

All checks run before every merge; each member works on a separate branch and merges through a pull request.

## 4. Plan for the remaining time (until the final submission)

| # | Task | Done when |
|---|---|---|
| 1 | Fix the story-mode money ending: compute it from story choices and build the emergency fund in story mode, so good play ends visibly better. | Simulator: good > random > bad on net worth and emergency fund. |
| 2 | Add the missing late-life lessons to story mode that the simulator flags. | Simulator: 24 / 24 lessons seen in every policy. |
| 3 | Turn the pre/post quiz into the learning measure, shown in the end-of-life lesson report for teachers and NGOs. | Report shows per-lesson score before and after. |
| 4 | Stretch: bring Sahukar's chair and Parallel Life replay into story mode. | Reachable from a story ending. |
| 5 | Full bug sweep: Hindi and Telugu text overflow on small screens, then re-run all tests. | All checks green on the final commit. |
| 6 | Submission: README (setup, run and third-party credits only), MIT licence, final report, demo video, code zip without dependencies, slides. | All deliverables in place before 11:00 AM. |

**Risks.** Balance changes can shift other outcomes, so we will change one number at a time and re-run the simulator after each. We will confirm the licence of the background music track before release, and replace it if needed.